

1. Prove convergence and calculate the integral

$$I = \int_0^{+\infty} x^3 e^{-x} dx.$$

Solution: We will prove the convergence calculating the integral by parts:

$$\begin{aligned} I &= \int_0^{+\infty} x^3 d(-e^{-x}) = -x^3 e^{-x} \Big|_0^{+\infty} + 3 \int_0^{+\infty} x^2 e^{-x} dx = 3 \int_0^{+\infty} x^2 e^{-x} dx = \\ &= -3x^2 e^{-x} \Big|_0^{+\infty} + 6 \int_0^{+\infty} x e^{-x} dx = 6 \int_0^{+\infty} x e^{-x} dx = -6x e^{-x} \Big|_0^{+\infty} + 6 \int_0^{+\infty} e^{-x} dx = -6e^{-x} \Big|_0^{+\infty} = 6. \end{aligned}$$

2. Prove convergence and find the sum of the series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k(k+1)}.$$

Solution: The integral converges by Leibniz test since $\frac{1}{k(k+1)}$ is decreasing and tends to 0. Consider power series

$$f(x) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1} x^{k+1}}{k(k+1)}.$$

The radius of convergence is equal to 1. Consequently,

$$f'(x) = \sum_{k=1}^{\infty} \frac{(-1)^{k-1} x^k}{k} = \ln(1+x)$$

and

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k(k+1)} = \int_0^1 f'(x) dx = \int_0^1 \ln(1+x) dx = (x+1) \ln(1+x) - x \Big|_0^1 = 2 \ln(2) - 1.$$

3. Check the convergence of the series

$$\sum_{k=1}^{\infty} \sin \frac{(\sqrt{k}+1)^3}{k^3 + 2k + 1}.$$

Solution:

$$\sin \frac{(\sqrt{k}+1)^3}{k^3 + 2k + 1} \sim \frac{(\sqrt{k}+1)^3}{k^3 + 2k + 1} \sim \frac{1}{k^{3/2}}.$$

Consequently the series converge since $\sum_{k=1}^{\infty} \frac{1}{k^{3/2}}$ converges.

4. Check the convergence of the improper integral

$$I = \int_0^{\infty} \frac{\sin \sqrt{x}}{x} dx.$$

Solution: Let $t = \sqrt{x}$. Then $dx = 2t dt$ and

$$I = 2 \int_0^{\infty} \frac{\sin t}{t} dx.$$

Consequently the series converges by Dirichlet's test since

$$\int \sin t dt = -\cos t + c$$

is bounded and the function $g(t) = \frac{1}{t}$ is monotonic and tends to 0.

5. Find the length of a path

$$x = \sin t, \quad y = \sin 2t, \quad t \in [0, 2\pi].$$

Solution: First,

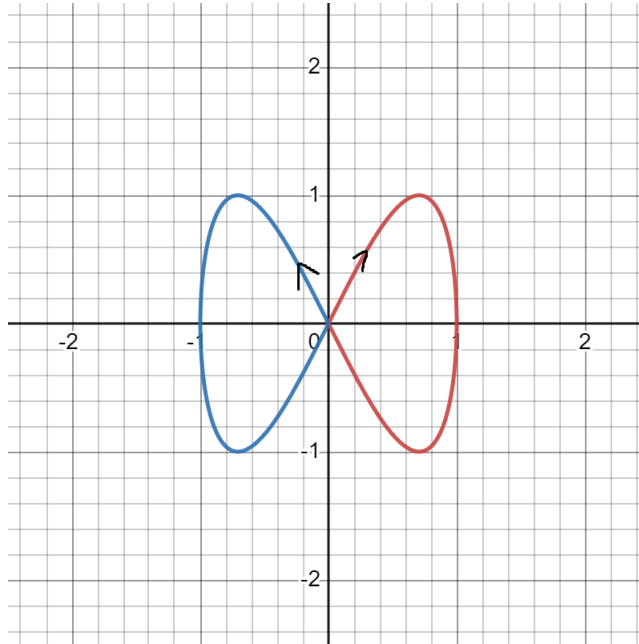
$$x'^2 + y'^2 = \cos^2 t + 2 \cos^2 2t = \frac{1 + \cos 2t + 4 \cos^2 2t}{2}$$

$$l(\gamma) = \frac{1}{\sqrt{2}} \int_0^{2\pi} \sqrt{1 + \cos 2t + 4 \cos^2 2t} dt.$$

It seems that I did a mistake in the statement of the problem. Find the area of a figure D bounded by the curve

$$x = \sin t, \quad y = \sin 2t, \quad t \in [0, 2\pi].$$

First, consider the sketch of the curve:



Notice that the domain is symmetric with respect to y -axis and the right part of the curve is oriented clockwise:

$$\begin{aligned} \text{Area}(D) &= 2 \int_0^{\pi} x(t)(-y'(t))dt = -2 \int_0^{\pi} \sin t \cos 2t dt = -2 \int_0^{\pi} \sin t (2 \cos^2 t - 1) dt = \\ &= \frac{4}{3} \cos^3 t \Big|_0^{\pi} - 2 \cos t \Big|_0^{\pi} = 4 - 8/3 = \frac{4}{3} \end{aligned}$$

Remark! The integral $\int_0^{2\pi} x(t)y'(t)dt = 0$. This integral doesn't take into account the orientation of the curve. Right and the left parts of the curve have opposite orientation relative the figure D .

6. Find the radius of convergence and the sum of the power series

$$f(x) = \sum_{k=1}^{\infty} \frac{2k}{2k+1} x^{2k+1}.$$

Solution: First,

$$R^{-1} = \lim_{k \rightarrow \infty} \sqrt[2k+1]{\frac{2k}{2k+1}} = 1.$$

$$f(x) = \sum_{k=1}^{\infty} x^{2k+1} - \sum_{k=1}^{\infty} \frac{x^{2k+1}}{2k+1} = \frac{x^3}{1-x^2} + x - \sum_{k=0}^{\infty} \frac{x^{2k+1}}{2k+1}.$$

Let $g(x) = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{2k+1}$. Then

$$g'(x) = \sum_{k=0}^{\infty} x^{2k} = \frac{1}{1-x^2}.$$

Consequently,

$$g(x) = g(0) + \int_0^x \frac{dt}{1-t^2} = \frac{1}{2} \ln \frac{1+x}{1-x}$$

and

$$f(x) = \frac{1}{1-x^2} - \frac{1}{2} \ln \frac{1+x}{1-x}.$$

7. Calculate the integral

$$\int_0^1 \ln(1-x)dx.$$

Solution: Let's find the indefinite integral first:

$$\int \ln(1-x)dx = x \ln(1-x) + \int \frac{x dx}{1-x} = x \ln(1-x) - \ln(1-x) = (x-1) \ln(1-x) - x.$$

Consequently,

$$\int_0^1 \ln(1-x)dx = ((x-1) \ln(1-x) - x) \Big|_0^1 = -1.$$

8. Let $p > 0$. Prove that

$$\lim_{n \rightarrow \infty} \int_n^{n+p} \frac{\sin^2 x}{x} dx = 0.$$

Solution: Using Mean value theorem we see that

$$x_n = \int_n^{n+p} \frac{\sin^2 x}{x} dx = \sin^2(c) \int_n^{n+p} \frac{dx}{x} = \sin^2(c) \ln \frac{n+p}{n}$$

for some $c \in [n, n+p]$. Consequently,

$$|x_n| \leq \ln \frac{n+p}{n} \rightarrow 0, \quad n \rightarrow \infty.$$

9. Determine without evaluating which integral is greater

$$I_1 = \int_0^{\pi} e^{-x^2} \sin^2 x dx \quad \text{or} \quad I_2 = \int_{\pi}^{2\pi} e^{-x^2} \sin^2 x dx.$$

Solution: Let $t = x - \pi$. Then $\sin x = -\sin t$ and

$$I_2 = \int_0^{\pi} e^{-(t+\pi)^2} \sin^2 t dt < I_1$$

since $e^{-(x+\pi)^2} < e^{-x^2}$ for every $x \in (0, \pi)$.

10. Check the uniform convergence of the series

$$\sum_{k=1}^{\infty} \frac{1}{n} \ln \frac{n+x}{n}$$

on $E_1 = (0, 1)$ and $E_2 = (1, +\infty)$.

Solution: Let $u_n(x) = \frac{1}{n} \ln \frac{n+x}{n}$ then

$$\sup_{x \in E_2} u_n(x) = +\infty$$

consequently, the series is not uniformly convergent on E_2 since $u_n \not\rightarrow 0$ on E_2 . Also

$$0 \leq \frac{1}{n} \ln \frac{n+x}{n} \leq \frac{1}{n} \ln \frac{n+1}{n} \leq \frac{1}{n^2}, \quad x \in (0, 1),$$

and the series converges uniformly by Weierstrass M-test and convergence of a series $\sum_{k=1}^{\infty} \frac{1}{k^2}$.

11. Prove the D'Alembert's ratio test for convergence of a series.

12. Check convergence and absolute convergence of a series

$$\sum_{k=1}^{\infty} \frac{\sin^3 k}{k}.$$

Solution:

$$\sin^3 k = \frac{3 \sin k - \sin 3k}{4}.$$

and series $\sum_{k=1}^{\infty} \frac{\sin kx}{k}$ is convergent for every x . Consequently, the series $\sum_{k=1}^{\infty} \frac{\sin^3 k}{k}$ is convergent.

To check the absolute convergence notice that

$$|\sin^3 k| \geq \sin^4 k = \frac{1}{4}(1 - \cos 2k)^2 = \frac{1}{4}(1 + \cos^2 2k - 2 \cos 2k) = \frac{3 - 2 \cos 2k + \cos 4k}{4}.$$

Series $\sum_{k=1}^{\infty} \frac{\cos ak}{k}$ is convergent when $\frac{a}{2\pi} \notin \mathbb{Z}$. The series $\sum_{k=1}^{\infty} \frac{1}{k}$ is divergent. Consequently, $\sum_{k=1}^{\infty} \frac{|\sin k|^3}{k}$ is divergent.

13. Find $\lim x_n$, where

$$x_n = \frac{1}{\sqrt{4n^2 - 1^2}} + \frac{1}{\sqrt{4n^2 - 2^2}} + \dots + \frac{1}{\sqrt{4n^2 - n^2}}.$$

Solution: Notice that

$$x_n = \frac{1}{n} \left(\frac{1}{\sqrt{4 - 1^2/n^2}} + \frac{1}{\sqrt{4 - 2^2/n^2}} + \dots + \frac{1}{\sqrt{4 - n^2/n^2}} \right) = \frac{1}{n} \sum_{k=1}^n \frac{1}{\sqrt{4 - \frac{k^2}{n^2}}}$$

is the integral sum for function $f(x) = \frac{1}{\sqrt{4-x^2}}$ and partition $\tau_n = \left\{ \frac{k}{n} \right\}_{k=0}^n$ of $[0, 1]$ with tags $\{\xi_k\}_{k=1}^n = \left\{ \frac{k}{n} \right\}_{k=0}^n$. The mesh of this partitions λ_{τ_n} tends to 0. Consequently,

$$\lim x_n = \int_0^1 \frac{dx}{\sqrt{4-x^2}} = [x = 2t] = \int_0^{1/2} \frac{dt}{\sqrt{1-t^2}} = \arcsin t \Big|_0^{1/2} = \arcsin \frac{1}{2} = \frac{\pi}{6}.$$

14. Prove that the absolutely convergent numerical series is convergent.

15. Prove the comparison test for improper integrals.

16. Prove the formula for integration by parts for Riemann integral.

17. Find the area of a curvilinear sector D bounded by a curve defined in polar coordinates

$$r(t) = 2 + \cos t, \quad t \in [0, 2\pi].$$

Solution:

$$\begin{aligned} \text{Area}(D) &= \frac{1}{2} \int_0^{2\pi} (2 + \cos t)^2 dt = \frac{1}{2} \int_0^{2\pi} (4 + 4 \cos t + \cos^2 t) dt = \\ &= \frac{1}{2} \int_0^{2\pi} \left(4 + 4 \cos t + \frac{1 + \cos 2t}{2} \right) dt = \frac{1}{4} (9 + 8 \sin t + \frac{1}{2} \sin 2t) \Big|_0^{2\pi} = \frac{9\pi}{2}. \end{aligned}$$